Tail bounds for Markov equivalence classes of uniform random DAGs

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Abstract

Let G_n be uniform among the simple directed acyclic graphs on n labeled vertices. We prove that its Markov equivalence class satisfies

$$\mathbb{P}\big(|\mathcal{M}(G_n)| > t\big) \le C \exp\left[-c\left(\frac{\log t}{\log\log t}\right)^2\right]$$

for all sufficiently large t, uniformly in n. This implies the upper tail with exponent $2-\varepsilon$ asked for by Schmid and Sly. The main estimate is a Gaussian tail for the number of vertices incident to reversible edges. Within a large essential-graph component, absence of unshielded colliders is costly; between many small components, their common-parent condition supplies the remaining quadratic cost. Source-layer enumeration transfers these estimates from independent forward edges to the uniform-DAG measure. Complete skeletons show that the logarithmic scale of the bound is sharp up to constants in the exponent.

1 Introduction and results

Let $\mathcal{D}_n$ be the set of simple directed acyclic graphs (DAGs) on $[n] = \{1, \dots, n\}$, and let G_n be uniform on $\mathcal{D}_n$. The skeleton of a DAG forgets its edge orientations. An unshielded collider, or v-structure, is a triple $u \to v \leftarrow w$ with u, w nonadjacent. We write

$$\mathcal{M}(G) = \{H \in \mathcal{D}_n : H \text{ has the same skeleton and unshielded colliders as } G\}.$$

These are the Markov equivalent DAGs; see [1, Theorem 2.1]. All logarithms below are natural.

Schmid and Sly [2, Theorem 1.1] prove a uniform upper tail with power $1 + 1/20$ on $\log t$ and ask, immediately after that theorem, whether the power can approach two from below. We prove the following stronger bound. The distribution here is uniform over labeled DAGs. Choosing a uniform vertex order and then independent forward edges gives a different law.

Theorem 1.1. *There are universal constants $c, C > 0$ and $t_0 > e$ such that, for every integer $n \ge 1$ and every real $t \ge t_0$,*

$$\mathbb{P}(|\mathcal{M}(G_n)| > t) \le C \exp\left[-c\left(\frac{\log t}{\log\log t}\right)^2\right]. \tag{1}$$

Corollary 1.2. *For every $\varepsilon \in (0,1)$ there are $C_\varepsilon < \infty$ and $t_\varepsilon > 1$ such that, simultaneously for every integer $n \ge 1$ and every real $t \ge t_\varepsilon$,*

$$\mathbb{P}\big(|\mathcal{M}(G_n)| > t\big) \le C_\varepsilon \exp[-(\log t)^{2-\varepsilon}].$$

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The essential graph of G retains an arrow when all members of $\mathcal{M}(G)$ give it the same orientation, and replaces every other edge by an undirected edge. Let $C_1, \dots, C_k$ be its undirected connected components with at least two vertices, and set

$$R(G) = \sum_{i=1}^{k} |C_i|,$$

with $R(G) = 0$ when $k = 0$. Thus R counts the vertices incident to reversible edges. Our main combinatorial estimate is

Theorem 1.3. *There are universal constants $c, C > 0$ such that, for all integers $n, m \geq 1$,*

$$\mathbb{P}(R(G_n) \geq m) \leq Ce^{-cm^2}. \tag{2}$$

The factorial bound $|\mathcal{M}(G)| \leq R(G)!$ then gives Theorem 1.1. The proof uses the source-layer enumeration underlying the tower method of Schmid and Sly [2, Section 2, Lemma 3.1 and Proposition 3.2], with elementary bounds sufficient for our purpose. We give those bounds in full. The additional quadratic estimate uses both the internal structure of each essential component and the arrows between components.

2 Necessary conditions on essential components

We use the essential-graph characterization of Andersson, Madigan and Perlman [1, Propositions 4.1–4.2 and Theorem 4.1]. The essential graph is a chain graph: it has no semidirected cycle, meaning a cycle containing at least one arrow whose arrows all point consistently around the cycle. Its undirected components are chordal, and each member of the equivalence class orients each component perfectly, that is, acyclically and without an unshielded collider inside the component. The essential graph also has no induced configuration $u \to v — w$ with u, w nonadjacent. The quoted locations are printed pages 10–11 of the primary-author technical report [1]; perfect orientations are defined in its Appendix A.

Lemma 2.1. *Every induced graph $G[C_i]$ has no unshielded collider. The directed graph obtained by contracting all essential components is acyclic. If $u \notin C_i$ and $u \to v$ for some $v \in C_i$, then $u \to w$ for every $w \in C_i$. Moreover,*

$$|\mathcal{M}(G)| \leq \prod_{i=1}^{k} |C_i|! \leq R(G)!. \tag{3}$$

Every reversible edge is absent from all unshielded colliders of G.

Proof. The internal assertion follows from perfect orientation. A directed cycle after contraction would yield a semidirected cycle before contraction, using undirected paths inside the components. For the parent assertion, take an undirected edge $v — w$ within C_i. The absence of an induced $u \to v — w$ forces u, w to be adjacent. Their edge is directed in the essential graph because u is outside the component. It cannot point from w to u, since $u \to v — w \to u$ would be a semidirected cycle. Hence $u \to w$. Propagation along undirected paths proves the assertion for every w.

All arrows between components are fixed throughout the class. An acyclic orientation of a component is determined by some topological order, and each order determines at most one orientation of its fixed skeleton. There are at most $|C_i|!$ such orientations. The product bounds the class size; the second inequality in (3) is the multinomial inequality. When $R = 0$ both sides equal one. Finally, either arrow of a specified unshielded collider must keep its direction throughout the class, so it cannot be reversible. □

3 Source layers and comparison with independent edges

Delete all sources of a DAG, then all sources of the remaining graph, and continue. Write the resulting nonempty layers as $H_1, \dots, H_T$, with $h_i = |H_i|$, and put $h_i = 0$ for $i > T$. The process terminates because every nonempty finite DAG has a source. Edges point from an earlier to a later layer, and every vertex of H_{i+1} has at least one parent in H_i. Conversely these two conditions characterize the source layers.

Conditionally on the graph, order each layer uniformly and independently, and concatenate the orders. We call the positions in this list ranks. This auxiliary randomness does not change the law of G_n. Let F_q be the induced graph on its first q ranks. For comparison, $\mathbb{Q}_q$ denotes the law on $[q]$ which includes each possible arrow $a \to b$, $a < b$, independently with probability $1/2$. An exact forward graph has $\mathbb{Q}_q$-mass $2^{-\binom{q}{2}}$.

Put $a_n = |\mathcal{D}_n|$, $a_0 = 1$, and

$$B_n = \frac{a_n}{n! 2^{\binom{n}{2}}}.$$

For a positive composition $h = (h_1, \dots, h_T)$ of n, define

$$W(h) = \prod_{i=1}^{T} \frac{2^{-\binom{h_i}{2}}}{h_i!} \prod_{i<T} (1 - 2^{-h_i})^{h_{i+1}}. \tag{4}$$

The empty composition has weight one.

Lemma 3.1. *We have $B_n = \sum_{|h|=n} W(h)$ and $\mathbb{P}(h(G_n) = h) = W(h)/B_n$. Conditionally on the ordered labeled layers, all edges joining layers at distance at least two are mutually independent fair bits. This remains true after the intralayer orders are chosen.*

Proof. The labeled layer sets can be chosen in $n!/\prod_i h_i!$ ways. Adjacent layers have $(2^{h_i} - 1)^{h_{i+1}}$ possible edge sets, since each vertex of the later layer needs a parent in the earlier one. Nonadjacent layers have unrestricted edges. The total is therefore

$$\frac{n!}{\prod_i h_i!} 2^{\sum_{i<j} h_i h_j} \prod_{i<T} (1 - 2^{-h_i})^{h_{i+1}}.$$

Divide by $n! 2^{\binom{n}{2}}$ and use $\binom{n}{2} = \sum_i \binom{h_i}{2} + \sum_{i<j} h_i h_j$. No adjacent-layer restriction involves a nonadjacent-layer edge, which also proves independence. The auxiliary orders are chosen independently of these edge bits. □

Lemma 3.2. *For every $n \geq 1$, every layer index $i \geq 1$, and every integer $x \geq 32$,*

$$\mathbb{P}(h_i \geq x) \leq 4\, 2^{-x^2/4}. \tag{5}$$

Proof. Fix $y \geq x$. Replace an entry $h_i = y$ by y consecutive ones, obtaining a composition $\widetilde{h}$ of the same n. For fixed i, y this map is injective, since the specified singleton entries can be collapsed again. Let $s = h_{i+1}$, taking $s = 0$ at the last layer. The preceding interface, if present, contributes a ratio at most one. The new singleton interfaces contribute 2^{y-1} to $W(h)/W(\widetilde{h})$, and the following interface contributes at most 2^s. Thus

$$\frac{W(h)}{W(\widetilde{h})} \leq \frac{2^{-\binom{y-1}{2}+s}}{y!}.$$

If $s \le x^2/16$, then $y \ge x \ge 32$ gives

$$-\binom{y-1}{2} + s \le -\frac{y^2}{2} + \frac{3y}{2} - 1 + \frac{y^2}{16} \le -\frac{y^2}{4}.$$

Summing over the injective image and dividing by B_n yields $\mathbb{P}(h_i = y, h_{i+1} \le x^2/16) \le 2^{-y^2/4}$. Consecutive terms in the sum over $y \ge x$ have ratio less than $1/2$, so

$$\mathbb{P}(h_i \ge x) \le \mathbb{P}\left(h_{i+1} \ge \lfloor x^2/16 \rfloor + 1\right) + 2\,2^{-x^2/4}.$$

Iterate with $x_0 = x$ and $x_{j+1} = \lfloor x_j^2/16 \rfloor + 1$. Then $x_{j+1} \ge 2x_j$ for $x_j \ge 32$. Once $x_j > n$ the remaining probability vanishes. The finite sum of error terms is bounded by $4\,2^{-x^2/4}$. □

Lemma 3.3. *For $n \ge 1$ and $0 \le q \le n$,*

$$B_n \ge 2^{-32} B_{n-1}, \qquad B_{n-q}/B_n \le 2^{32q}. \tag{6}$$

Proof. Take a DAG on $n-1$ labels, with S sources. Choose one of the n labels for a new vertex, connect it to every old source, and choose arbitrary outgoing edges to the other $n-1-S$ vertices. The new vertex is the unique source. Deleting it recovers the input and its chosen edges, so this is an injective count, after the remaining labels are identified in increasing order with $[n-1]$. Hence

$$a_n \ge n 2^{n-1} a_{n-1} \mathbb{E}[2^{-S}], \qquad B_n \ge B_{n-1} \mathbb{E}[2^{-S}].$$

Lemma 3.2 gives $\mathbb{P}(S \le 31) \ge 1/2$ and thus $\mathbb{E}[2^{-S}] \ge 2^{-32}$. For $n = 1$ the empty input has $S = 0$. Iteration proves the second inequality. □

Call q a layer boundary if it is the sum of an initial sequence of layer sizes; n is always a boundary.

Lemma 3.4. *For every exact forward graph F on $[q]$, where $1 \le q \le n$,*

$$\mathbb{P}(q \text{ is a boundary}, F_q = F) \le 2^{32q} \mathbb{Q}_q(F). \tag{7}$$

Consequently, for every integer $L \ge 32$, $\ell = \min(n, L)$, and every event A of forward graphs on $[\ell]$,

$$\mathbb{P}(F_\ell \in A) \le 2L\, 2^{64L} \mathbb{Q}_\ell(A) + 4L\, 2^{-L^2/4}. \tag{8}$$

Proof. An admissible complete rank graph J has contiguous source-layer blocks of sizes h. Each permutation from ranks to original labels gives a pair consisting of a labeled DAG and its chosen intralayer orders. All $n!$ pairs are distinct, including when the DAG has automorphisms, because the orders are part of the pair. Each has probability $1/(a_n \prod_i h_i!)$. Therefore

$$\mathbb{P}(F_n = J) = \frac{2^{-\binom{n}{2}}}{B_n \prod_i h_i!}.$$

If q is a boundary, the source layers of F are contiguous blocks; their vector $g = (g_1, \dots, g_r)$ is uniquely determined by F. The source layers of the induced remainder are exactly the remaining layers of the full graph. Cross edges are free except that every vertex of the first remainder layer

needs a parent in the last prefix layer. Summing over remainder graphs and cross edges gives the exact identity

$$\mathbb{P}(q \text{ boundary}, F_q = F) = \frac{2^{-\binom{q}{2}}}{B_n \prod_{i=1}^{r} g_i!} \sum_{|h|=n-q} W(h)(1-2^{-g_r})^{h_1}. \tag{9}$$

When $q = n$, the sum and its interface factor are both one. Otherwise the sum is at most B_{n-q}. Lemma 3.3 proves (7). If F has noncontiguous source layers, the event is empty; there is no additional sum over prefix vectors.

For (8), if $n < L$ use the terminal boundary $q = n$. If $n \geq L$, take the first boundary $q \geq L$. When $q > 2L$, the layer containing rank L has size at least L, and its index is at most L. Lemma 3.2 and a union bound give probability at most $4L\,2^{-L^2/4}$ for this exception. Otherwise sum (7) over $L \leq q \leq \min(2L, n)$, restricting each graph to its first L ranks. Under $\mathbb{Q}_q$ that restriction has law $\mathbb{Q}_L$. There are at most $L + 1 \leq 2L$ boundaries to consider, and $2^{32q} \leq 2^{64L}$. □

We call an edge non-collider if it belongs to no unshielded collider.

Lemma 3.5. *There are universal $c_1, C_1 > 0$ such that, for all integers $n, L \geq 1$,*

$$\mathbb{P}(\textit{some non-collider edge has an endpoint of rank} > L) \leq C_1 e^{-c_1 L}. \tag{10}$$

Proof. For a possible larger endpoint $b \geq 6400$, let $r = \lfloor b/100 \rfloor$. By Lemma 3.2, the chance that one of the first b layers has size greater than r is at most $4b\,2^{-r^2/4}$. Condition on ordered layers satisfying this size cutoff. An edge $a \to b$, $a < b$, must join distinct layers. Among the first $\lfloor b/2 \rfloor$ ranks, discard those in layers at distance at most one from either endpoint's layer. At most $6r$ ranks are discarded, leaving at least $b/3$ witnesses. Every layer meeting these earlier ranks has index at most b; a layer after the layer of b contains none of them.

For each remaining witness u, the bits $\{u, b\}$ and $\{u, a\}$ are independent fair bits by Lemma 3.1. These pairs are also disjoint across witnesses. Since $u < b$, the first edge, if present, points to b. With probability $1/4$ it is present and $\{u, a\}$ is absent. If $a \to b$ is present, this gives $a \to b \leftarrow u$. Thus the event that $a \to b$ exists and is non-collider has conditional probability at most $(3/4)^{b/3}$. We have not conditioned on the existence of $a \to b$, so adjacent-layer restrictions on that edge cause no problem.

A union bound over $a < b$ gives

$$b(3/4)^{b/3} + 4b\,2^{-\lfloor b/100 \rfloor^2/4}.$$

Since $\lfloor b/100 \rfloor \geq b/200$, this is at most Ae^{-cb} for universal positive constants. Sum over $b > L$, extending the sum to infinity to remove dependence on n. Increasing C_1 covers $L < 6400$. □

4 Quadratic cost of the component conditions

We first work under the independent-edge law $\mathbb{Q}$.

Lemma 4.1. *There are universal $a, A > 0$ such that, for every integer $r \geq 2$,*

$$\mathbb{Q}_r(\textit{no unshielded collider}) \leq e^{-ar^2 + Ar}. \tag{11}$$

Proof. For $r \geq 16$ let U be the first $\lfloor r/2 \rfloor$ vertices, V the remaining vertices, and $k = \lfloor r/8 \rfloor \geq r/16$. Reveal only the edges within U. If the graph of missing skeleton edges on U has a matching of size k, choose one by a deterministic rule. For each $v \in V$, absence of colliders requires that v not receive both endpoints of any matched pair. The parent bits are distinct and unrevealed. Hence the conditional probability is at most

$$(3/4)^{k|V|} \leq \exp[-\log(4/3)\, r^2/32].$$

If no such matching exists, a maximum matching leaves at least $\lfloor r/2 \rfloor - 2(k-1) \geq r/4 - 1$ unmatched vertices. No missing edge joins two of them. They therefore form a clique of size at least k. The probability of this alternative is at most

$$2^r 2^{-\binom{k}{2}} \leq \exp[-(\log 2) r^2/512 + (33/32)(\log 2) r],$$

where $\binom{k}{2} \geq r^2/512 - r/32$ follows from $k \geq r/16$. Adding the two alternatives proves (11) with, for example,

$$a = \min\{\log(4/3)/32, \log 2/512\}, \qquad A = 2 \log 2.$$

For $2 \leq r < 16$ the right side of (11) exceeds one. □

Select s vertices of a $\mathbb{Q}_\ell$ graph, partition them into blocks $B_1, \ldots, B_k$ of sizes $r_i \geq 2$, and fix a proposed topological order of the blocks. Require each induced block to have no unshielded collider. For $i < j$, require each vertex of B_i to be a parent either of all vertices of B_j or of none, and forbid arrows from B_j to B_i. These are necessary conditions from Lemma 2.1 for distinct essential components, or for subsets of them.

Lemma 4.2. *For a fixed selected set, partition and block order, the probability of these conditions is at most*

$$\exp(-a_0 s^2 + As), \qquad a_0 = \min\{a, \log 2/4\} > 0. \tag{12}$$

Let $A_{m,\ell}$ be the event that some selected set of size $m \leq s \leq 2m$ admits such a partition and order. There is a universal $d > 0$ such that, for every sufficiently large integer m and $\ell \leq L \leq m^2$,

$$\mathbb{Q}_\ell(A_{m,\ell}) \leq e^{-dm^2}. \tag{13}$$

Proof. For a fixed $u \in B_i$, $i < j$, at most two patterns on its r_j pairs with B_j are allowed: all arrows $u \to B_j$, or all pairs absent. If a rank in B_j precedes u, the all-parent pattern is impossible, which only decreases the count. The interblock probability is therefore at most $2^{-r_i(r_j-1)}$. Since $r_j \geq 2$,

$$\sum_{i<j} r_i(r_j - 1) \geq \tfrac{1}{2} \sum_{i<j} r_i r_j = \tfrac{1}{4} \left(s^2 - \sum_i r_i^2 \right).$$

Internal and interblock edge sets are disjoint. The induced law within each selected block is $\mathbb{Q}_{r_i}$, regardless of gaps between its ranks. Lemma 4.1 thus bounds the full probability by

$$\exp\left[-a \sum_i r_i^2 + As - \frac{\log 2}{4} \left(s^2 - \sum_i r_i^2 \right) \right] \leq e^{-a_0 s^2 + As}.$$

There are at most L^s choices of vertices, s^s partitions and $s!$ block orders. For $m \leq s \leq 2m$, $L \leq m^2$, the logarithm of this overcount is at most $s(4 \log m + 2 \log 2)$. Together with As, it is at most $(a_0/2)s^2$ for all sufficiently large m. Sum the resulting bound $e^{-a_0 m^2/2}$ over at most $m + 1$ values of s. One may take $d = a_0/4$ after increasing the threshold. Values $s > \ell$ give empty events. □

5 Proof of the tails and their scale

Proof of Theorem 1.3. Let $D = 64\log 2 + 1$, choose $\kappa = d/(4D) < 1$, and put $L = \lfloor \kappa m^2 \rfloor$. For large enough m, we have $32 \le L \le m^2$ and $L \ge \kappa m^2/2$. If $n < m$ there is nothing to prove. Otherwise, on $R \ge m$, choose m vertices from one component of size at least m, if such a component exists. If all components are smaller, accumulate whole nontrivial components until the total first reaches m. Their total size s then lies in $[m, 2m)$. In either case the selected blocks have at least two vertices and inherit the conditions of Lemma 2.1. They need not be essential components of the induced prefix graph.

By Lemma 2.1, every vertex counted by R is incident to a non-collider edge. Outside the exception of Lemma 3.5, all such vertices have rank at most L. Thus the selection witnesses $A_{m,\ell}$ in F_ℓ, where $\ell = \min(n, L)$. Lemmas 3.4 and 4.2 give

$$\mathbb{P}(R \ge m) \le C_1 e^{-c_1 L} + 2L\, 2^{64L} e^{-dm^2} + 4L\, 2^{-L^2/4}. \tag{14}$$

Since $\log(2L) \le L$, the middle term is at most $\exp(DL - dm^2) \le e^{-3dm^2/4}$. The first term is at most $C_1 e^{-c_1 \kappa m^2/2}$. The last is bounded by $C' e^{-c' m^4}$, and hence by a Gaussian in m. All constants are independent of n. Increasing C covers the finitely many smaller positive integers m. □

Proof of Theorem 1.1 and Corollary 1.2. Write $x = \log t$ and, for large enough x, put

$$m = \lfloor x/(2\log x) \rfloor.$$

Then $m \ge x/(4\log x)$, while

$$\log(m!) \le m \log m \le m \log x \le x/2.$$

In particular $m! \le t$. By (3), $|\mathcal{M}(G_n)| > t$ implies $R(G_n) > m$. Theorem 1.3 now proves (1), after changing its universal constant c. For each fixed $\varepsilon \in (0,1)$, $x^\varepsilon/(\log x)^2 \to \infty$. Choose X_ε above the preceding thresholds so that $cx^\varepsilon/(\log x)^2 \ge 1$ for all $x \ge X_\varepsilon$. With $t_\varepsilon = e^{X_\varepsilon}$ and $C_\varepsilon = C$, (1) gives Corollary 1.2. □

Proposition 5.1. *The Gaussian scale in (2) and the scale $(\log t/\log\log t)^2$ in (1) are sharp up to universal constants in the exponent along complete-skeleton examples. In particular, no constants $c, C > 0$ give a uniform upper bound $C\exp[-c(\log t)^2]$ for the class-size tail at all large t.*

Proof. Every DAG has a topological order, so $a_n \le n! 2^{\binom{n}{2}}$. Exactly $n!$ DAGs have complete skeleton, one for each total order. They belong to one class, of size $n!$, and have $R = n$ for $n \ge 2$. Consequently

$$\mathbb{P}(\text{complete skeleton}) \ge 2^{-\binom{n}{2}}.$$

Also $n\log n - n + 1 \le \log(n!) \le n\log n$, by integrating the increasing function $\log x$. For $t_n = n! - 1$ we have $\log t_n \sim n\log n$ and $\log t_n/\log\log t_n \sim n$. The displayed lower bound thus has the same exponential scales as (2) and (1). A proposed bound $Ce^{-c(\log t)^2}$ at $t = t_n$ would imply $c(\log t_n)^2 \le \log C + (\log 2)\binom{n}{2}$, which is impossible after division by n^2. □

Finite check. A standalone exact-enumeration script accompanies the manuscript. For four vertices it enumerates all 543 labeled DAGs and their 185 Markov equivalence classes. It checks the source-layer weights, the exact prefix identity (9), the common-parent condition and the factorial bound using integer and rational arithmetic. These finite checks are ancillary; no computation is used as a premise of the proofs above.

References

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[2] D. Schmid and A. Sly, *On the number and size of Markov equivalence classes of random directed acyclic graphs*, arXiv:2209.04395v2 (2024), revised September 28, 2024.

AI use. OpenAI Codex (GPT-6.1 Sol) was used to develop and check the argument, verify source passages, implement the supplementary finite enumeration, and prepare this manuscript. The manuscript has not been formally verified or independently reviewed by a human. Independent mathematical review is requested.